🎯 Quiz: Model Fine-Tuning & LLM Optimization

🧠 Section 1: Conceptual MCQs

1. **What is model fine-tuning?**
   A. Training a model from scratch
   B. Adjusting a pre-trained model on new data
   C. Only changing prompts
   D. Increasing model size

   Ans. B

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2. **Which scenario is best suited for prompting instead of fine-tuning?**
   A. Domain-specific chatbot
   B. Repetitive structured output task
   C. One-time query with general knowledge
   D. Training on private dataset

   Ans. C

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3. **What is the main advantage of fine-tuning over prompting?**
   A. Faster inference always
   B. Better control over task-specific outputs
   C. No need for data
   D. Lower memory usage

   Ans. B

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4. **What does LoRA stand for?**
   A. Low Rank Adaptation

B. Linear Representation Adjustment
C. Layer Optimization and Reuse Algorithm
D. Learning Rate Adjustment

Ans. A

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5. **In LoRA, what is actually trained?**
   A. Entire model weights
   B. Only input tokens
   C. Small low-rank matrices
   D. Output layer only

   Ans. C

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6. **Why is LoRA considered parameter-efficient?**
   A. It reduces dataset size
   B. It updates only a small subset of parameters
   C. It removes layers
   D. It compresses input

   Ans. B

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7. **What is the role of a tokenizer in fine-tuning?**
   A. Train model weights
   B. Convert text into tokens
   C. Evaluate model
   D. Store embeddings

   Ans. B

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8. **Which step comes first in a fine-tuning workflow?**
   A. Model evaluation
   B. Data collection & cleaning
   C. Training
   D. Deployment

   Ans. B

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9. **What is overfitting in fine-tuning?**
   A. Model performs well on all data
   B. Model memorizes training data but fails on new data
   C. Model ignores training data
   D. Model becomes faster

   Ans. B

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10. **Which metric is commonly used for classification tasks?**
    A. BLEU score
    B. Accuracy
    C. Perplexity
    D. Latency

    Ans. B

🔍 Section 2: True / False

11. Fine-tuning always requires large datasets. FALSE
12. Prompting does not change model weights. TRUE
13. LoRA modifies all layers of the model. FALSE
14. Validation data is used to tune hyperparameters. TRUE
15. Fine-tuned models can outperform base models on niche tasks. TRUE

🧩 Section 3: Scenario-Based Questions

16. Scenario:
You need a chatbot specialized in legal documents with consistent tone and formatting.
👉 What should you choose?
A. Prompting
B. Fine-tuning
C. Zero-shot learning
D. Random initialization

Ans. B

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17. Scenario:
You want quick results for generating marketing slogans without training.
👉 Best approach?
A. Fine-tuning
B. Prompt engineering
C. LoRA
D. Training from scratch

Ans. B

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18. Scenario:
You have limited GPU resources but want to adapt a large model.
👉 Best method?
A. Full fine-tuning
B. LoRA
C. Increase batch size
D. Add more layers

Ans. B

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19. Scenario:
Your model performs well on training data but poorly on test data.
👉 What is the issue?
A. Underfitting
B. Overfitting
C. Data leakage
D. Tokenization error

Ans. B

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20. Scenario:
You are comparing two models for summarization.
👉 Which metric is most relevant?
A. Accuracy
B. BLEU / ROUGE
C. Latency only
D. Token count

Ans. B

⚙️ Section 4: Short Answer

21. Explain one difference between prompting and fine-tuning.

Ans 21. □ Prompting: You guide the model using input text (no training).

□ Fine-tuning: You train the model on new data to change its behavior.

22. What is one benefit of using LoRA?

Ans 22. LoRA reduces memory and training cost by updating only small parts of the model instead of the full model.

23. List any two steps in a fine-tuning workflow.

Ans 23. □ Prepare and clean the dataset

□ Train the model on labeled data

24. What is the purpose of validation data?

Ans. To check model performance during training and prevent overfitting.

25. Name one metric used to evaluate LLMs.

Ans. Accuracy
(Other examples: F1-score, BLEU, Perplexity)

🧪 Section 5: Practical Thinking

26. You are fine-tuning a model and loss is not decreasing. What could be one reason?

Ans. Learning rate may be too high or too low

27. Why might fine-tuning be expensive?

Ans. Requires large datasets, powerful GPUs, and long training time

28. When should you avoid fine-tuning?

Ans. □ When simple prompting already gives good results

□ When you have very little data

29. What happens if tokenizer is mismatched with the model?

Ans. Input is incorrectly processed → model gives poor or wrong output

30. How does LoRA reduce computational cost?

Ans. Input is incorrectly processed → model gives poor or wrong output